STUDENTNEST

COMP 3613 Assignment 2

Software testing plans, execution results, and evidence

Student Accommodation Application

Student	**Malik Blackman**
Student ID	**816045522**

Project: StudentNest, a student accommodation listing, management, and review application.
Submission: Assignment 2 only. The Assignment 1 report remains a separate artifact.
Prepared: November 2026.

Project and test scope

StudentNest is a FastAPI and SQLModel web application. Students browse and search listings, view listing details, create bookings, and submit reviews for eligible listings. Landlords manage their own listings. The user directory is restricted to administrators.

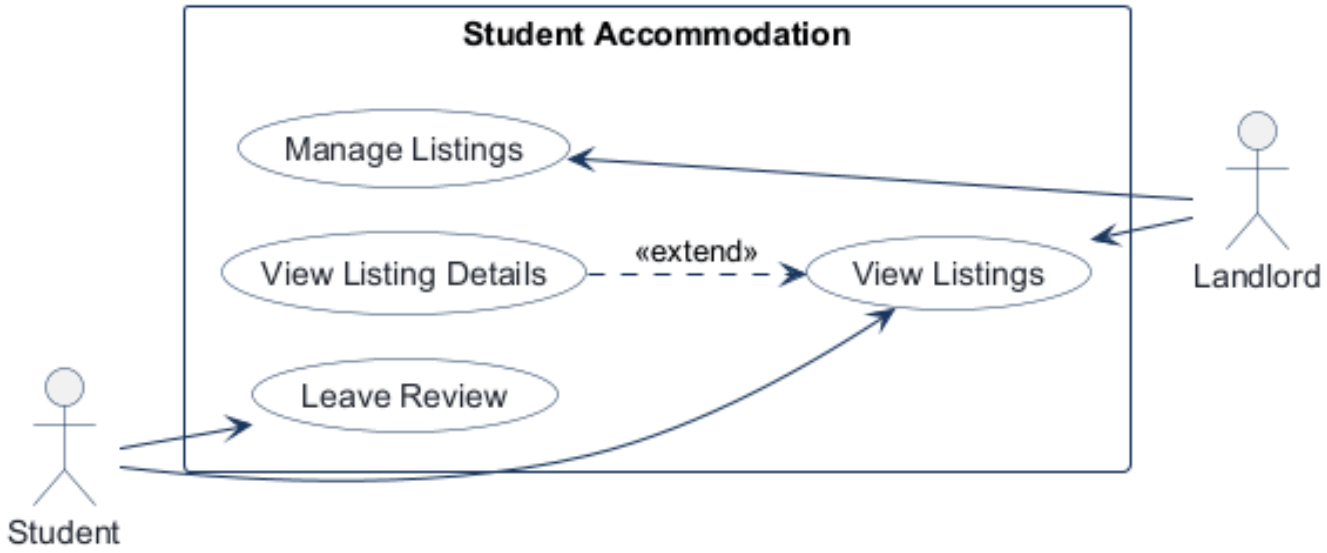
Workflows under test

View Listings	Public listing browse/search and listing detail for students.
Manage Listings	Authenticated landlord listing creation and owner-scoped management.
Leave Review	A student with a prior booking submits and views a listing review.

Test setup

Automated API tests use a fresh in-memory SQLite database per test. The developer and Render databases are not used by the test suite. Manual UAT was performed in the local web application; Postman exercised the API at http://127.0.0.1:5000. The collection is included in the repository at postman/studentnest-api.postman_collection.json.

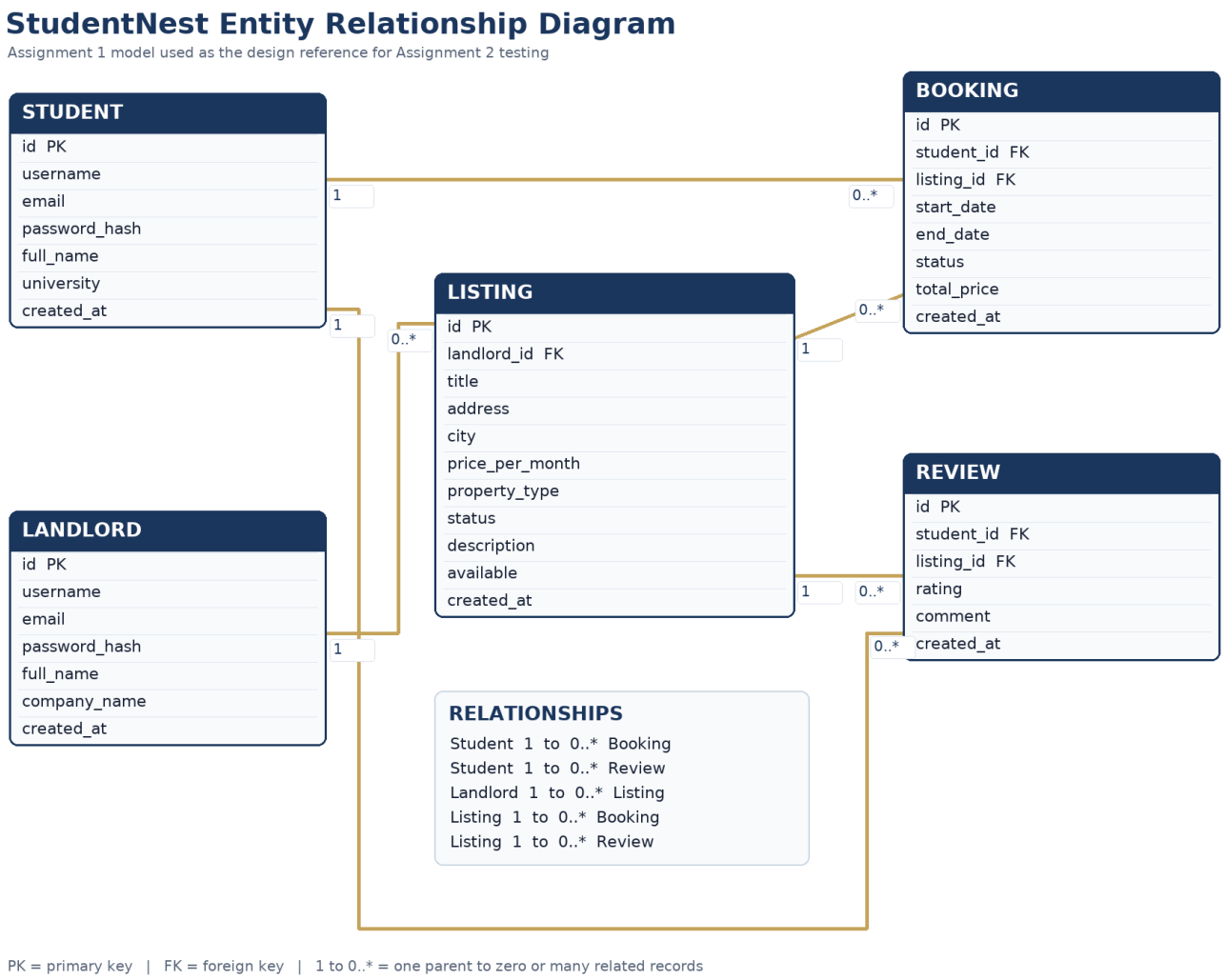
Use-case diagram



Reused as Assignment 1 design context so the Assignment 2 test scope is traceable to the named workflows.

Data model reference

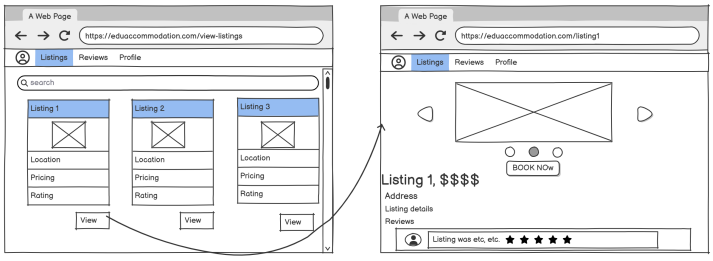
This ERD reproduces the StudentNest model used as the design reference for the Assignment 2 tests: Student, Landlord, Listing, Booking, and Review.



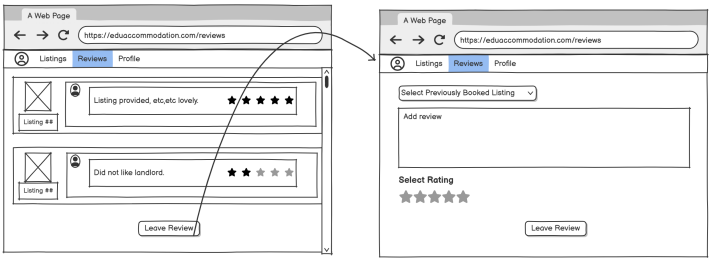
Application wireframe reference

The student-crafted Assignment 1 wireframe connects the manual acceptance tests to the intended listing, review, and landlord management screens.

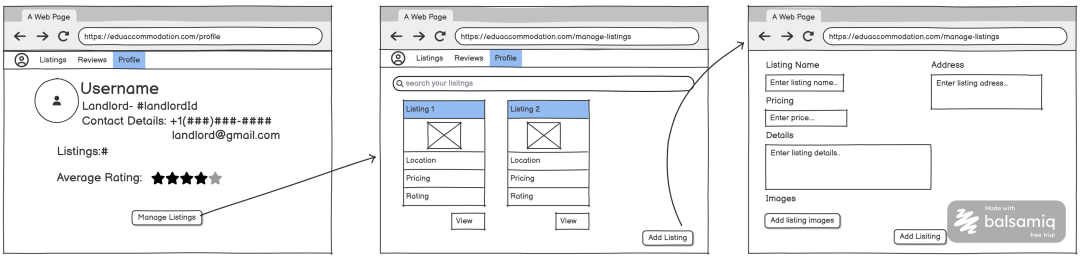
View Listing



Leave Review



Manage Listings



User acceptance testing

Each UAT case was performed in the browser and compared with the relevant workflow and visible application state. A passing UI restriction does not replace the corresponding direct API authorization test.

UAT-01 View Listings / Student

Search for a matching listing, open its detail, and perform a no-match search.
PARTIAL: no-match search shows the expected empty state; detail view exposes a literal rating-template placeholder.

UAT-02 Manage Listings / Landlord

Search the current landlord?s management list for another landlord?s listing.
PASS: ?palm residences? is not shown; the management empty state appears.

UAT-03 Leave Review / Student

Submit a rating and comment for a previously booked listing.
PASS: confirmation appears and the submitted review is shown.

UAT-04 Review eligibility / Student

Inspect listing choices in the review form.
PASS (UI flow): only previously booked listings are offered; direct API rejection is tested separately.

UAT-05 Signed-out access

Inspect the signed-out navigation and available actions.
PASS (UI flow): public listings and sign-in/register are visible; unauthenticated API rejection is tested separately.

Manual test environment

The recorded local UAT environment was started with the project?s documented initialization and run commands. UAT-01 remains open as a rendering defect.

Automated test plan and results

Unit tests isolate AccommodationService behavior with mocked repositories. Integration tests exercise HTTP routing, validation, authentication, service/repository interaction, persistence, and response schemas using an isolated in-memory SQLite database.

Unit coverage

UT-01 Listing creation normalization and defaults

UT-02 Booking monthly price calculation

UT-03 Reject same-day and reversed booking dates

UT-04 Reject reviews without a prior booking

UT-05 Create a review and trim the comment

UT-06 Aggregate landlord ratings across owned listings

UT-07 Return no average when no reviews exist

Integration/API coverage

IT-01 Public listing search and detail response

IT-02 Unknown listing returns 404

IT-03 Anonymous listing creation rejected; authenticated creation succeeds

IT-04 Landlord listing results scoped to the current owner

IT-05 Booking/review authentication and booking eligibility

IT-06 Non-increasing booking dates rejected without persistence

IT-07 User directory restricted to administrators; passwords omitted

Observed results

Latest combined pytest run: 15 passed (8 unit + 7 integration), with 19 deprecation warnings and no failures. Individual captured runs show 8 passed for unit tests and 7 passed for integration tests. Warnings are associated with the existing Starlette/httpx test-client combination and naive datetime defaults.

Pytest execution evidence

The screenshots show the separately captured unit and integration test runs. A later combined run recorded 15 passed overall.

Unit tests ? 8 passed

```
PS C:\Users\super\Documents\GitHub\comp3613a1> python -m pytest -q tests/test_accommodation_service.py
........                                                                 [100%]
============================================= warnings summary =============================================
..\..\..\AppData\Roaming\Python\Python313\site-packages\fastapi\testclient.py:1
  C:\Users\super\AppData\Roaming\Python\Python313\site-packages\fastapi\testclient.py:1: StarletteDeprecationWarnin
g: Using `httpx` with `starlette.testclient` is deprecated; install `httpx2` instead.
    from starlette.testclient import TestClient as TestClient  # noqa

tests/test_accommodation_service.py: 11 warnings
  C:\Users\super\AppData\Roaming\Python\Python313\site-packages\pydantic\_internal\_fields.py:881: DeprecationWarni
ng: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version. Use timezone-aware obje
cts to represent datetimes in UTC: datetime.datetime.now(datetime.UTC).
    return fac()

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
8 passed, 12 warnings in 0.14s
PS C:\Users\super\Documents\GitHub\comp3613a1>
```

Integration tests ? 7 passed

```
.......                                                                  [100%]
============================================= warnings summary =============================================
..\..\..\AppData\Roaming\Python\Python313\site-packages\fastapi\testclient.py:1
  C:\Users\super\AppData\Roaming\Python\Python313\site-packages\fastapi\testclient.py:1: StarletteDeprecationWarnin
g: Using `httpx` with `starlette.testclient` is deprecated; install `httpx2` instead.
    from starlette.testclient import TestClient as TestClient  # noqa

tests/test_accommodation_api.py::test_public_listing_list_search_and_detail
tests/test_accommodation_api.py::test_listing_creation_requires_auth_and_records_owner
tests/test_accommodation_api.py::test_landlord_listing_collection_is_scoped_to_current_user
tests/test_accommodation_api.py::test_landlord_listing_collection_is_scoped_to_current_user
tests/test_accommodation_api.py::test_booking_and_review_api_enforce_auth_and_booking_rule
tests/test_accommodation_api.py::test_booking_and_review_api_enforce_auth_and_booking_rule
tests/test_accommodation_api.py::test_booking_rejects_non_increasing_dates
  C:\Users\super\AppData\Roaming\Python\Python313\site-packages\pydantic\_internal\_fields.py:881: DeprecationWarni
ng: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version. Use timezone-aware obje
cts to represent datetimes in UTC: datetime.datetime.now(datetime.UTC).
    return fac()

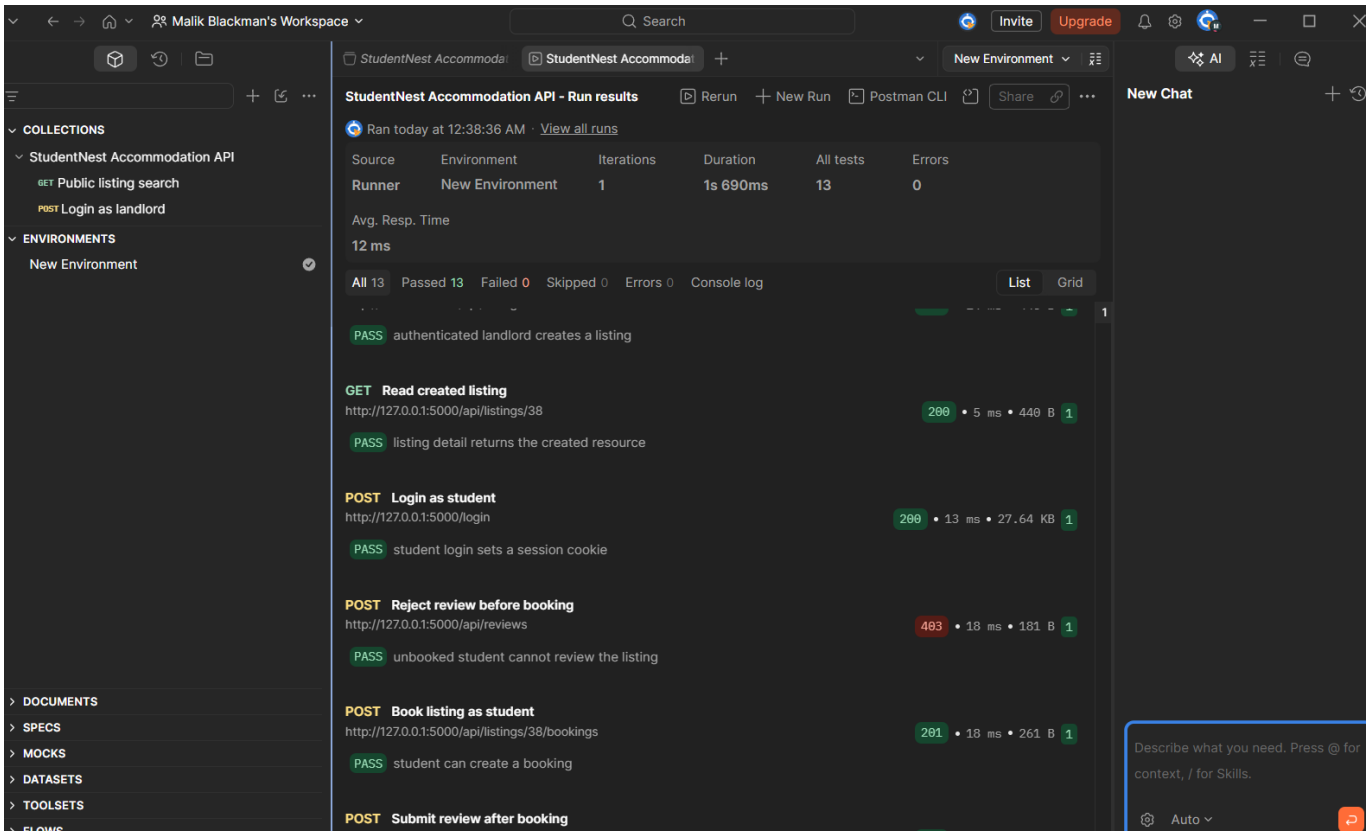
-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
7 passed, 8 warnings in 0.33s
PS C:\Users\super\Documents\GitHub\comp3613a1>
```

Postman API collection

The first collection run recorded 10 passed and 3 failed because Postman followed successful login redirects and returned the final HTTP 200 response while the assertions expected only 302/303. The collection was updated to accept 200, 302, or 303 for login. Subsequent authenticated requests confirmed session cookies were used.

Successful rerun ? 13 of 13 checks passed

Recorded result: 13 tests, 13 passed, 0 failed, 0 skipped, 0 errors. Requests cover public search, landlord listing creation, student booking/review checks, and admin-only user access.

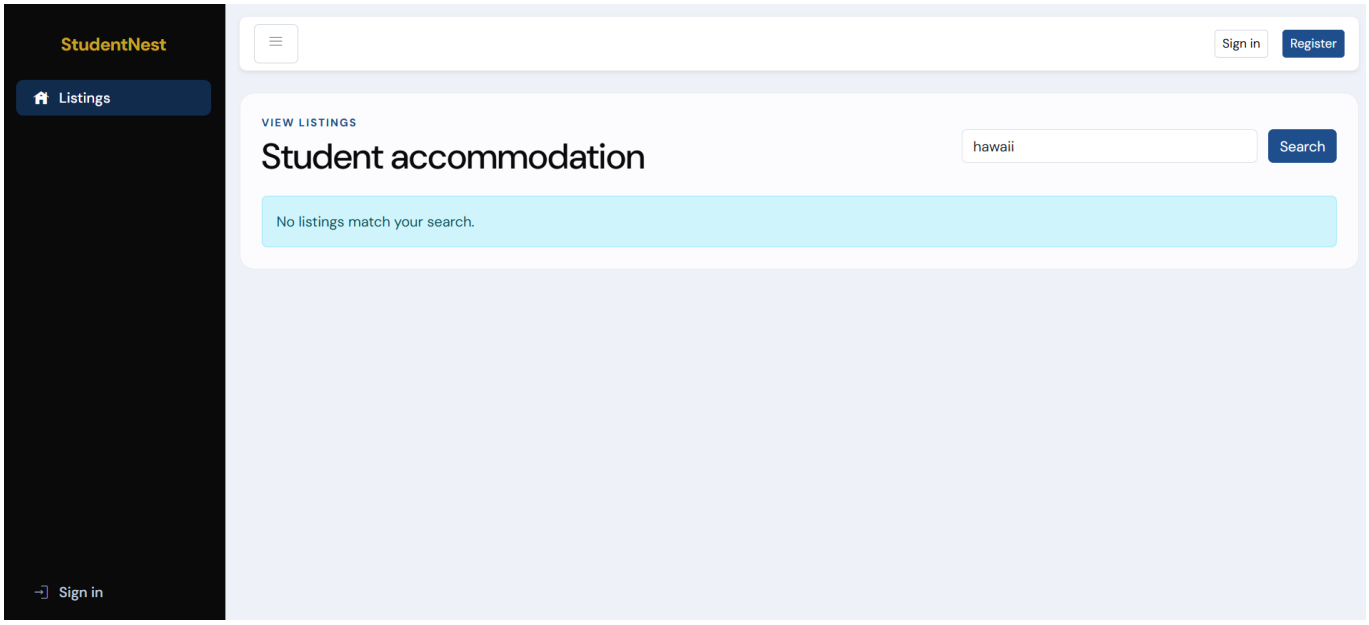


Collection: postman/studentnest-api.postman_collection.json

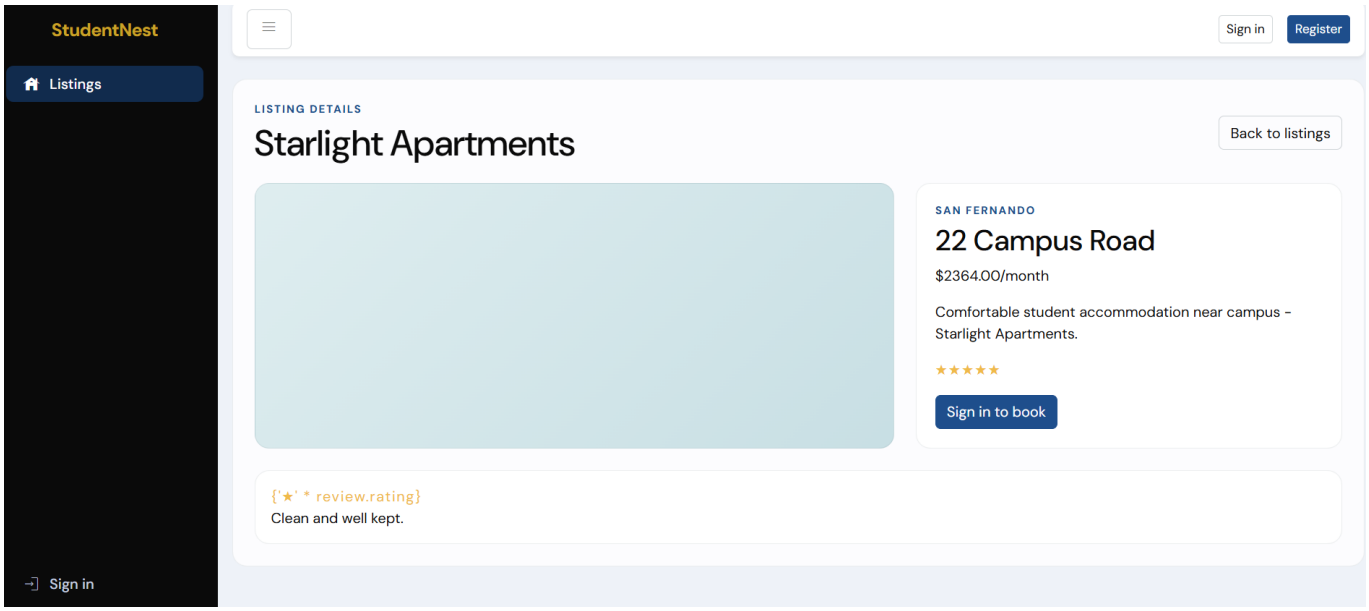
UAT-01 evidence ? partial

The no-match search behaves as expected. The listing detail screenshot shows literal ?{?* review.rating}? text instead of rendered rating stars, so UAT-01 is not a full pass.

No-match search state



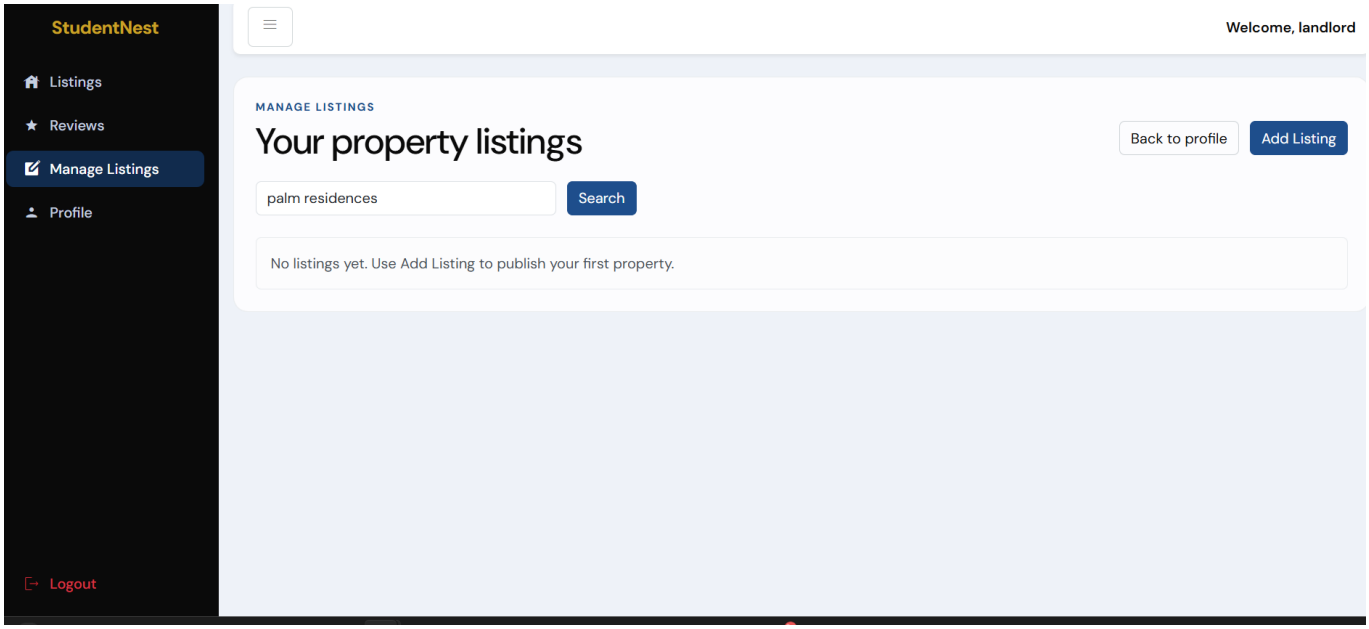
Listing detail rendering defect



UAT-02 and UAT-03 evidence

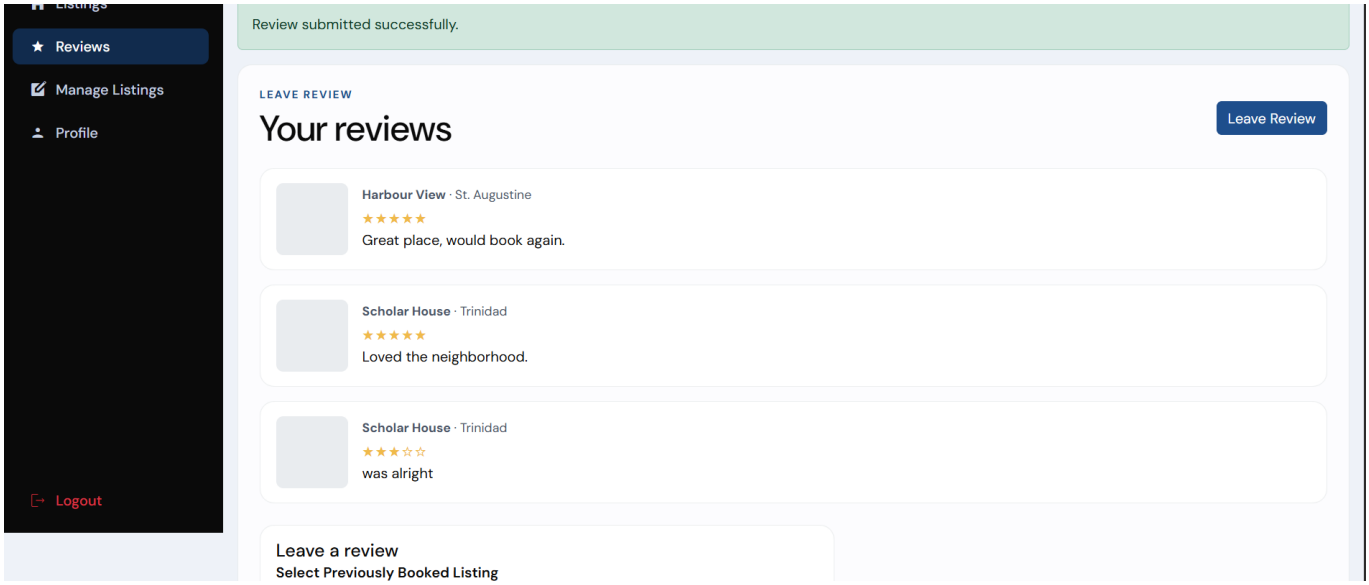
UAT-02 ? landlord owner-scoped management

Searching Manage Listings for ?palm residences? returned the no-listings state for the signed-in landlord, consistent with results being limited to that account?s listings.



UAT-03 ? review submission

After submission, the application displayed a success message and the new review in the student?s review list.



Submission references and status

Source artifacts

Assignment 2 test plans: docs/assignment2/test-plans.md
Assignment 2 submission record: docs/assignment2/submission.md
Unit tests: tests/test_accommodation_service.py
Integration tests: tests/test_accommodation_api.py
Postman collection: postman/studentnest-api.postman_collection.json

Project references

Repository: https://github.com/malikBlackman99/comp3613a1
Public application: https://faststarter-0g4k.onrender.com
Assignment 1 report: docs/report.md (reference only; not replaced or merged).

Final result

Automated test evidence is passing: 15 pytest cases and 13 Postman checks. UAT-02 through UAT-05 are recorded as passes. UAT-01 remains partial because of the rating rendering defect, despite the expected no-results search state. This package preserves that limitation.